

**Correct Answer: C**

**Rationale**

Choice C is the best answer. The graph shows that the CNO-treated females were way more active than the CNO-treated males, while the saline-treated males and females (the control groups) had very similar activity levels. This supports the claim that there were sex-specific differences in the mice's response to neuron activation.

Choice A is incorrect. This choice misreads the graph. All four groups of mice started at nearly the same activity level before treatment (see how all four points are very close together at -4 minutes, meaning four minutes before treatment). Choice B is incorrect. This choice doesn't complete the assertion. The assertion is about the mice's response to neuron activation, so we need to include the data about the CNO-treated females and males. Choice D is incorrect. This choice misreads the graph. The line for the CNO-treated males does not show a "substantial decline" until around 122 minutes, and there is no "steep increase" afterward.